# CHE 251

# (Term Project)

# FINAL REPORT

**(Group 1)**


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# INVESTIGATING HYDROTREATING PROCESS OPTIMIZATION THROUGH HYDROGEN PARTIAL PRESSURE MANAGEMENT WITH $H_2$ RECYCLE

# Contents

# 1 Project Objectives and Scope

This project aims to develop a comprehensive, simulation-based framework to model a heavy diesel hydrotreating unit. The core objectives for this final report are focused on not only establishing a robust mathematical model for the base case operation but also on optimizing this process to achieve enhanced results through advanced analysis.

- To construct and validate a DWSIM flowsheet of the hydrotreating process, complete with a hydrogen recycle loop, using kinetic models from established literature and Chao-Seader VLE correlations.
- To establish a complete and mathematically closed set of governing equations (mass balances, energy balances, and kinetic laws) for each unit operation in the base case.
- To perform a detailed Degrees of Freedom (DOF) analysis to formally prove that the model is well-defined and solvable.
- To perform a multi-objective Pareto analysis to identify optimal operating conditions that maximize sulfur conversion while minimizing total energy consumption.
- To define and analyze an "increased efficiency case" based on the Pareto front, demonstrating enhanced output and comparing its performance against the base case.

This report presents the complete project, from model development and validation to optimization and the final analysis of an increased efficiency case leading to enhanced output.

# 2 Process Flowsheet

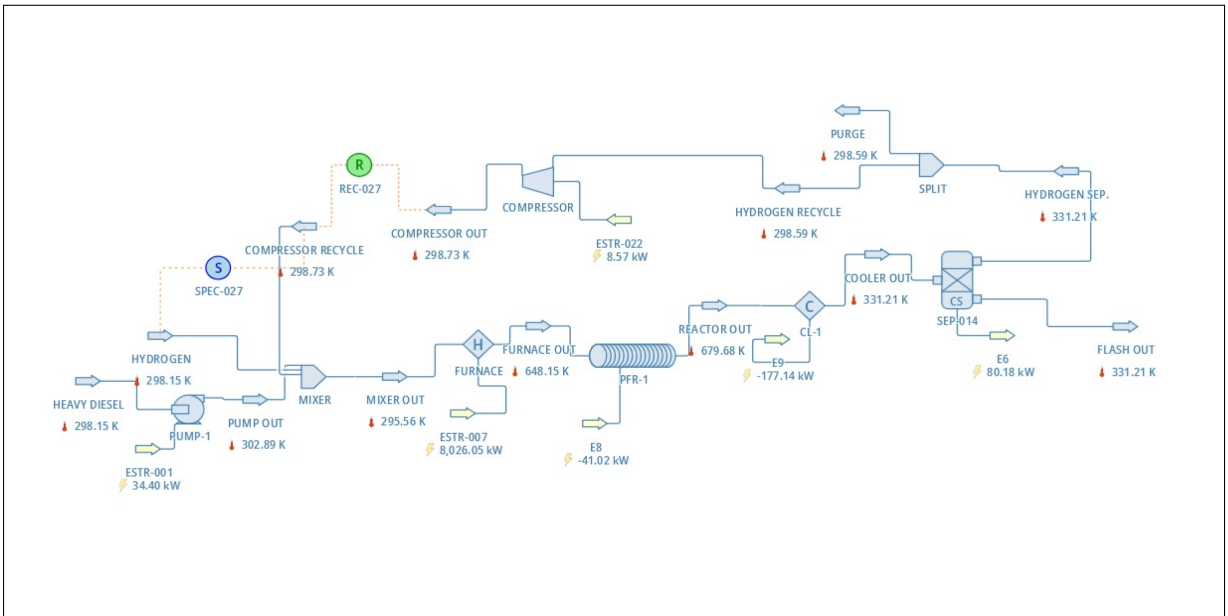


Figure 1: Process Flow Diagram of the Hydrotreating Unit with Hydrogen Recycle Loop.

# 3 Necessary Information for Solution

## 3.1 Mass and Energy Balances in Context

- **Mass Balance:** For any unit operation at steady state, the rate of mass entering must equal the rate of mass leaving. For reactive systems like the PFR, this principle is extended to component molar balances, where the outlet flow of a species equals its inlet flow plus the amount generated or minus the amount consumed by the reaction.
- **Energy Balance:** The first law of thermodynamics dictates that energy is conserved. For an open system at steady state, the total enthalpy of outlet streams must equal the total enthalpy of inlet streams plus any heat added ($\dot{Q}$) and minus any work done by the system ($\dot{W}$).

## 3.2 Main Chemical Reaction

The core of the process is the catalytic hydrodesulfurization (HDS) of sulfur compounds, represented by the reaction of methanethiol:

$$\mathrm{CH_3SH + H_2 \longrightarrow CH_4 + H_2S} \tag{1}$$

## 3.3 Main Kinetics Equation

The rate of sulfur consumption ($-r_s$) is described by the power-law kinetic model from Buitrago et al. (2017):

$$-r_s = K \cdot C_S{}^{*} \cdot P_{H_2}^{0.4} \tag{2}$$

*

The rate constant, $K$, is temperature-dependent via the Arrhenius equation:

$$K = K_0 \exp\left(-\frac{E_a}{RT}\right) \tag{3}$$

# 4 Basis of Calculation

- **Time Basis:** 1 hour of steady-state operation.
- **Feed Stream:** Heavy Diesel inlet stream.
- **Feed Flow Rate:** Total molar flow is 36 kmol/h.
- **Feed Composition:** Contains **6 mol% sulfur**, modeled as methanethiol ($CH_3SH$).

# 5 Variables

## 5.1 Known Variables (Inputs and Specifications)

*The kinetic model was revised from $C_S{}^2$ to $C_S$ in the Base Case to align with literature recommendations for diesel hydrodesulfurization kinetics.

Table 1: Input Conditions

| Category | Variable | Symbol | Value |
|---|---|---|---|
| Initial Feed Stream | Total Molar Flow | $F_{Diesel}$ | 36 kmol/h |
| | Sulfur Concentration | $Z_{CH_3SH}$ | 6 mol% |
| | Inlet Temperature | $T_{Inlet}$ | 25 °C |
| | Inlet Pressure | $P_{Inlet}$ | 0.101325 MPa |
| Reactor Conditions | Operating Pressure | $P_{System}$ | 12.5 MPa |
| | Reaction Temperature | $T_{Rxn}$ | 375 °C |
| Design Constraints | $H_2$/Sulfur Ratio | - | 1000 (Molar) |
| | LHSV | - | 2.0 L/(h·$g_{cat}$) |
| | Efficiencies | $\eta_{Pump}, \eta_{Comp}$ | 75% |
| | Split Ratio (Recycle) | $S$ | 85% |
| | Cooler Outlet Temp | $T_{Cooler,out}$ | 58.06 °C |
| | Flash Sep. Pressure | $P_{Flash}$ | 12.5 MPa |
| | Makeup $H_2$ Temp | $T_{H_2,makeup}$ | 25 °C |
| Kinetic Parameters | Activation Energy | $E_a$ | 68.6 Kcal/mol |
| | Pre-exp. Factor | $\ln(K_0)$ | 59.068 |

†

## 5.2 Unknown Variables (Solved Outputs)

Table 2: Unknown Variables to be Solved by the Model

| Category | Variable | Symbol |
|---|---|---|
| Core Performance | Sulfur Conversion | $\chi_S$ (%) |
| Design Parameters | Reactor Volume | $V_{Reactor}$ ($m^3$) |
| Stream Flows | Makeup $H_2$ Flow | $F_{H_2,Makeup}$ |
| | Total & Component Flows | $F_{Total,j}$, $F_{i,j}$ |
| Stream Properties | Temperature & Enthalpy | $T_j$, $H_j$ |
| Energy Duties (ESTRs) | Pump Work | $\dot{W}_{Pump}$ (ESTR-001) |
| | Furnace Heat Duty | $\dot{Q}_{Furnace}$ (ESTR-007) |
| | Reactor Heat Duty | $\dot{Q}_{Reactor}$ (E8) |
| | Cooler Heat Duty | $\dot{Q}_{Cooler}$ (E9) |

† For simulating the defined process in DWSIM correctly we intended to change some of the values that we marked in red. These changes were made to increase efficiency and conversion for the process.

| Category | Variable | Symbol |
| --- | --- | --- |
| | Compressor Work | $\dot{W}_{Comp}$ (ESTR-022) |

The simulation tracks 11 chemical species ($N_c = 11$) across the flowsheet, categorized as follows:

- **A. Reacting Species** ($i$): These are the components involved in the hydrodesulfurization (HDS) reaction:
  - Methanethiol ($CH_3SH$)
  - Hydrogen ($H_2$)
  - Methane ($CH_4$)
  - Hydrogen Sulfide ($H_2S$)
- **B. Inert Hydrocarbon Species** ($i$): These components make up the bulk of the Heavy Diesel stream. They are non-reactive but their thermodynamic properties are crucial for VLE and heat balances:
  - n-Hexane
  - n-Decane
  - n-Hexadecane
  - Benzene
  - Toluene
  - P-xylene
  - Cyclohexane

# 6 Governing Equation Set by Unit Operation

## 6.1 Unit 1: PUMP-001 (Heavy Diesel Feed Pump)

- **Purpose:** To increase the pressure of the heavy diesel feed to the required process pressure.
- **Governing Equations:**

  **Component Mass Balance ($N_c$ equations):**

$$F_{in,i} = F_{out,i} \tag{4}$$

  **Energy Balance (1 equation):**

$$\dot{W}Pump = \frac{Fin(H_{out} - H_{in})}{\eta_{Pump}} \tag{5}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $N_c$ (for $F_{out,i}$) + 1 ($T_{out}$) + 1 ($H_{out}$) + 1 ($\dot{W}_{Pump}$) = $N_c + 3$
  - Number of Equations: $N_c$ (Mass Balances) + 1 (Energy Balance) + 1 (Property Relation $T = f(H,P)$) + 1 (Specification: $P_{out}$) = $N_c + 3$
  - **DOF = (Variables) - (Equations) = 0**

### 6.2 Unit 2: Mixer

- **Purpose:** To combine the pressurized diesel, makeup hydrogen, and recycle streams.

- **Governing Equations:**

  **Component Mass Balance ($N_c$ equations):**

$$F_{out,i} = F_{diesel,i} + F_{makeup,i} + F_{recycle,i} \tag{6}$$

  **Energy Balance (1 equation):**

$$F_{out} H_{out} = \sum_{j \in inlets} F_j H_j \tag{7}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $N_c$ (for $F_{out,i}$) + 1 ($T_{out}$) + 1 ($H_{out}$) = $N_c + 2$
  - Number of Equations: $N_c$ (Mass Balances) + 1 (Energy Balance) + 1 (Property Relation $T = f(H,P)$) = $N_c + 2$
  - **DOF = 0**

### 6.3 Unit 3: Furnace (Heater)

- **Purpose:** To heat the mixed feed to the isothermal reactor temperature of 375°C.

- **Governing Equations:**

  **Component Mass Balance ($N_c$ equations):**

$$F_{in,i} = F_{out,i} \tag{8}$$

  **Energy Balance (1 equation):**

$$\dot{Q}Furnace = Fin(H_{out} - H_{in}) \tag{9}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $N_c$ ($F_{out,i}$) + 1 ($H_{out}$) + 1 ($\dot{Q}_{Furnace}$) = $N_c + 2$
  - Number of Equations: $N_c$ (Mass Balances) + 1 (Energy Balance) + 1 (Specification: $T_{out} = 375°C$) = $N_c + 2$
  - **DOF = 0**

### 6.4 Unit 4: Plug Flow Reactor (PFR)

- **Purpose:** To carry out the catalytic HDS reaction.

- **Governing Equations:**

**Component Mass Balance ($N_c$ equations):**

$$F_{out,i} = F_{in,i} + \nu_i \cdot (\chi_S \cdot F_{in,CH_3SH}) \tag{10}$$

**Isothermal Energy Balance (1 equation):**

$$\dot{Q}Reactor = \sum iF_{out,i}H_{out,i} - \sum_i F_{in,i}H_{in,i} \tag{11}$$

**Design Equation (1 equation):**

$$V_{Reactor} = \frac{\dot{V}_{Liquid,Feed}}{LHSV} \tag{12}$$

**Kinetic Relation (1 equation):**

$$V_{Reactor} = F_{in,CH_3SH} \int_0^{\chi_S} \frac{d\chi'_S}{-r_s} \tag{13}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $N_c$ ($F_{out,i}$) + 1 ($\chi_S$) + 1 ($V_{Reactor}$) + 1 ($\dot{Q}_{Reactor}$) = $N_c + 3$
  - Number of Equations: $N_c$ (Mass Balances) + 1 (Energy Balance) + 1 (Design Eq.) + 1 (Kinetic Relation) = $N_c + 3$
  - **DOF = 0**

## 6.5 Unit 5: Cooler

- **Purpose:** To cool the reactor effluent for separation.
- **Governing Equations:**

**Component Mass Balance ($N_c$ equations):**

$$F_{in,i} = F_{out,i} \tag{14}$$

**Energy Balance (1 equation):**

$$\dot{Q}Cooler = Fin(H_{out} - H_{in}) \tag{15}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $N_c$ ($F_{out,i}$) + 1 ($H_{out}$) + 1 ($\dot{Q}_{Cooler}$) = $N_c + 2$
  - Number of Equations: $N_c$ (Mass Balances) + 1 (Energy Balance) + 1 (Specification: $T_{out}$) = $N_c + 2$
  - **DOF = 0**

## 6.6 Unit 6: SEP-014 (Flash Separator)

- **Purpose:** To separate the effluent into vapor (recycle) and liquid (product) streams.

- **Governing Equations:**

  **Component Mass Balance ($N_c$ equations):**

$$F_{feed,i} = F_{vapor,i} + F_{liquid,i} \tag{16}$$

  **Phase Equilibrium ($N_c$ equations):**

$$y_i = K_i x_i \tag{17}$$

  **Energy Balance (1 equation):**

$$F_{feed} H_{feed} = F_{vapor} H_{vapor} + F_{liquid} H_{liquid} \tag{18}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $2N_c$ (flows $F_{V,i}, F_{L,i}$) + 2 (T, P of outlets) + 2 ($H_V$, $H_L$) = $2N_c + 4$
  - Number of Equations: $N_c$ (Mass Balances) + $N_c$ (Phase Equilibria) + 1 (Energy Balance) + 1 (Rachford-Rice) + 2 (Property relations for H) = $2N_c + 4$
  - **DOF = 0**

## 6.7 Unit 7: Splitter

- **Purpose:** To divide the vapor stream into a purge and a recycle stream.
- **Governing Equations:**

  **Component Mass Balance ($N_c$ equations):**

$$F_{in,i} = F_{purge,i} + F_{recycle,i} \tag{19}$$

  **Split Ratio Definition ($N_c$ equations):**

$$F_{recycle,i} = S \cdot F_{in,i} \tag{20}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $2N_c$ (for $F_{purge,i}, F_{recycle,i}$) = $2N_c$
  - Number of Equations: $N_c$ (Mass Balances) + $N_c$ (Split Ratio Specs) = $2N_c$
  - **DOF = 0**

## 6.8 Unit 8: Compressor

- **Purpose:** To re-pressurize the recycle gas stream.
- **Governing Equations:**

**Component Mass Balance ($N_c$ equations):**

$$F_{in,i} = F_{out,i} \tag{21}$$

**Energy Balance (1 equation):**

$$\dot{W}Comp = \frac{Fin(H_{out} - H_{in})}{\eta_{Comp}} \tag{22}$$

- **Degrees of Freedom Analysis:**
  - Number of Variables: $N_c$ ($F_{out,i}$) + 1 ($T_{out}$) + 1 ($H_{out}$) + 1 ($\dot{W}_{Comp}$) = $N_c + 3$
  - Number of Equations: $N_c$ (Mass Balances) + 1 (Energy Balance) + 1 (Property Relation) + 1 (Spec: $P_{out}$) = $N_c + 3$
  - **DOF = 0**

## 6.9 DWSIM Model Construction and Base Case Solution‡

The comprehensive mathematical model was realized as a flowsheet in the DWSIM simulator, utilizing the specified Base Case conditions: $T_{Rxn} = 375$ °C, $P_{System} = 12.5$ MPa, and Split Ratio $S = 0.85$.

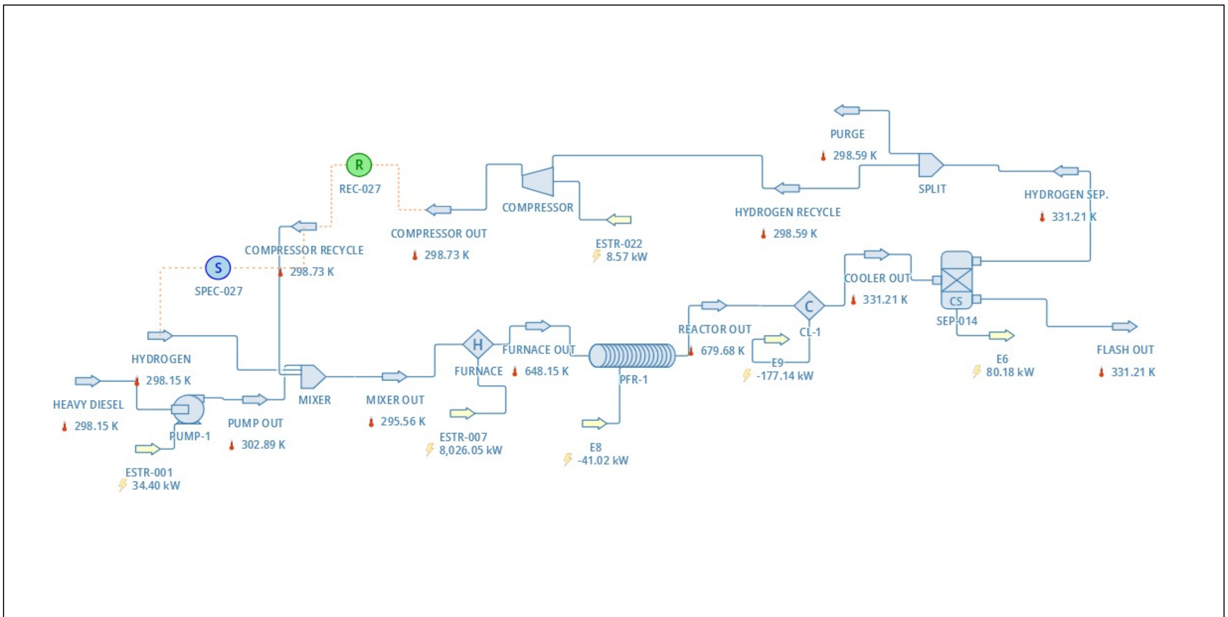


Figure 2: Process Flow Diagram of the Hydrotreating Unit with Hydrogen Recycle Loop.

‡

### Process Flow Description

The hydrotreating process involves eight key unit operations configured to maximize contact between the hydrocarbon feed and hydrogen. The process stream flow is detailed below:

‡From this page onwards every detail corresponds to our final term readings and calculations.

1. **Feed Preparation and Mixing (PUMP-1, MIXER OUT):**
   - The Heavy Diesel feed is first pressurized by PUMP-1 ($\dot{W} = 34.40$ kW) to the operating pressure.
   - This stream combines with the $H_2$ Makeup stream and the compressed Recycle gas in the Mixer.
2. **Preheating (FURNACE):**
   - The mixed stream is heated in the Furnace (H) ($\dot{Q} = 8,012.68$ kW) to a furnace out temperature of 375°C (648.15 K).
3. **Reaction (PFR-1):**
   - The heated feed enters the Plug Flow Reactor (PFR-1), where the Hydrodesulfurization (HDS) reaction occurs.
   - The reactor operates with a defined outlet temperature, requiring a minimal cooling duty ($\dot{Q} = -40.65$ kW) to remove the heat of reaction.
4. **Cooling and Separation (CL-1, SEP-014):**
   - The reactor effluent (REACTOR OUT, 679.69 K) is drastically cooled by Cooler (CL-1) ($\dot{Q} = -176.83$ kW) to 331.21 K.
   - The cooled stream enters the Flash Separator (SEP-014), yielding the final liquid product (FLASH OUT) and a hydrogen-rich vapor stream (HYDROGEN SEP.).
5. **Recycle Loop (SPLIT, COMPRESSOR):**
   - The vapor stream is divided by the Splitter (SPLIT). A portion is removed as PURGE (298.56 K) to control inert buildup.
   - The remaining HYDROGEN RECYCLE stream is pressurized by the COMPRESSOR ($\dot{W} = 8.47$ kW) and routed back to the Mixer, completing the loop.

**Base Case Performance Benchmark**

The converged solution using the flowsheet provided the following initial performance benchmark:

- **Base Case Sulfur Conversion ($\chi_S$)**: 99.03%
- **Base Case Total Energy Input ($\sum \dot{W} + \sum \dot{Q}$)**: $28,506.42$ kJ/h

## 6.10 MATLAB Validation and Thermodynamic Model Comparison

The governing equations were independently solved in MATLAB to verify the DWSIM base case and to investigate the influence of the chosen Thermodynamic Model on the final results. A models was implemented in MATLAB for the calculation of Vapor-Liquid Equilibrium (VLE): the Chao-Seader model (used in the DWSIM base case).

#### 6.10.1 Summary of Performance by Chao-Seader

Table 3: Summary Performance of the Model (Chao-Seader Results)

| Metric/Duty | Unit | Chao-Seader Results |
|---|---|---|
| **Core Performance** | | |
| Overall Conversion ($\chi_S^{ov}$) | % | 100 |
| **Design Parameters** | | |
| Catalyst Weight ($W_{cat}$) | kg | 33.70 |
| **Energy Duties (kW)** | | |
| Pump Work ($W_{pump}$) | kW | 32.48 |
| Furnace Duty ($Q_{furnace}$) | kW | 7091.00 |
| Reactor Duty ($Q_{reactor}$) | kW | -42.05 |
| Cooler Duty ($Q_{cooler}$) | kW | -6925.91 |
| Compressor Work ($W_{comp}$) | kW | 0.00 |

§

#### 6.10.2 Detailed Stream Results: Chao-Seader Model

The following tables present the detailed stream results and composition analysis for the Chao-Seader model, which is consistent with the thermodynamic model chosen for the DWSIM base case.

Table 4: Stream Properties - MATLAB (Chao-Seader)

| Stream | T (°C) | H (kJ/kmol) | Flow (kmol/h) |
|---|---|---|---|
| 1. Diesel Feed | 25.00 | 0.00 | 36 |
| 2. Pump Out | 35.46 | 3247.79 | 36 |
| 3. $H_2$ Makeup | 25.00 | 964.44 | 315.41 |
| 4. Recycle In | 58.06 | 2056.33 | 1792.01 |
| 5. Furnace Out | 375.00 | 1915.55 | 2212.13 |
| 6. Reactor Out | 375.00 | 13825.15 | 2143.32 |
| 7. Product (Liquid) | 58.06 | 10356.04 | 35.07 |

§The discrepancy between the 100% conversion calculated in MATLAB and the 99.03% conversion from DWSIM originates from the difference in modeling assumptions for the heterogeneous reactor. The custom MATLAB script calculates the intrinsic reaction rate based purely on the provided Arrhenius kinetics, effectively assuming an effectiveness factor ($\eta$) of 1.0. This ideal model presumes that every reactant molecule has immediate access to all catalyst active sites. Conversely, DWSIM accounts for the physical realities of a porous catalyst, recognizing that the reaction is **diffusion-limited**. Because the intrinsic chemical reaction is extremely fast at 375°C, the true bottleneck is the rate at which reactants can diffuse from the bulk fluid into the catalyst's pores. DWSIM calculates a realistic effectiveness factor ($\eta \ll 1$), where reactants are consumed on the pellet's exterior faster than they can penetrate the interior. This results in a slower observed reaction rate ($rate_{\textbf{observed}} = \eta \cdot rate_{\textbf{kinetic}}$) and, consequently, a more physically accurate conversion of 99.03%.

| Stream | T (°C) | H (kJ/kmol) | Flow (kmol/h) |
|---|---|---|---|
| 8. Purge (Vapor) | 58.06 | 2056.33 | 316.24 |

Table 5: Stream Flows & Compositions - MATLAB (Chao-Seader, kmol/h)

| Component | Feed | Product (Liquid) | Purge (Vapor) |
|---|---|---|---|
| $CH_3SH$ | 2.088 | 0.000 | 0.000 |
| $H_2$ | 0.000 | 0.434 | 312.820 |
| $CH_4$ | 0.000 | 0.003 | 2.052 |
| $H_2S$ | 0.000 | 1.095 | 0.992 |
| n-Hexane | 1.696 | 1.613 | 0.083 |
| n-Decane | 8.478 | 8.464 | 0.014 |
| n-Hexadecane | 13.565 | 13.565 | 0.000 |
| Benzene | 0.677 | 0.649 | 0.028 |
| Toluene | 2.714 | 2.676 | 0.039 |
| P-xylene | 1.696 | 1.687 | 0.008 |
| Cyclohexane | 5.087 | 4.884 | 0.202 |

# 7 Results and Discussion

## 7.1 Pareto Optimization and Trade-off Analysis: Methodology

To define the optimal operating point beyond the single Base Case, a multi-objective optimization was performed using the Pareto optimization method. This analysis maps the Pareto Front—the boundary of non-dominated solutions where performance in one objective cannot be improved without sacrificing another.

A comprehensive parameter sweep was conducted by varying four key Decision Variables: $T_{Rxn}$, LHSV, Recycle Ratio ($S$), and System Pressure ($P_{System}$). The analysis aimed to simultaneously Maximize Overall Sulfur Conversion and Minimize Capital/Operating Costs (Catalyst Weight and Total Power).

## 7.2 Analysis of Economic and Process Trade-offs

### 7.2.1 Raw Material and Capital Cost Analysis

**1. Raw Material Cost ($H_2$ Makeup)**: The Pareto front for $H_2$ consumption was nearly vertical (Figure 3). This is a critical finding: all optimal designs, across the full conversion range ($\sim$ 48% to 100%), consumed a **fixed amount of $H_2$ makeup** (approx. 13.5 kmol/h). This simplifies the optimization by treating $H_2$ makeup as a constant cost rather than a variable trade-off.

**2. Capital Cost (CAPEX) vs. Conversion**: This plot (Figure 4) clearly shows the trade-off between reactor size (CAPEX) and performance. Designs with LHSV = 3.0 represent the lowest capital cost

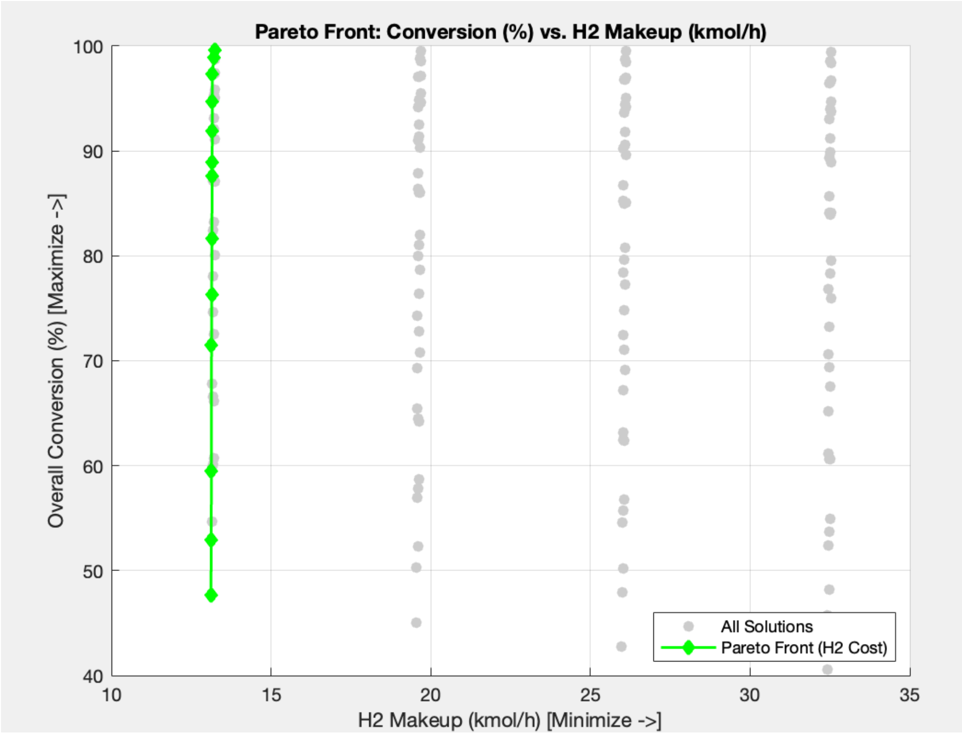


Figure 3: Pareto Front: Raw Material Cost ($H_2$ Makeup) vs. Overall Conversion.

($\sim 1.35$ kg). Critically, at this minimum CAPEX, a high conversion up to $\sim 95\%$ is achievable. Gaining a marginal $2-3\%$ more conversion requires jumping to the next cluster (LHSV = 2.5), demanding a significantly larger and more expensive reactor ($\sim 1.62$ kg).

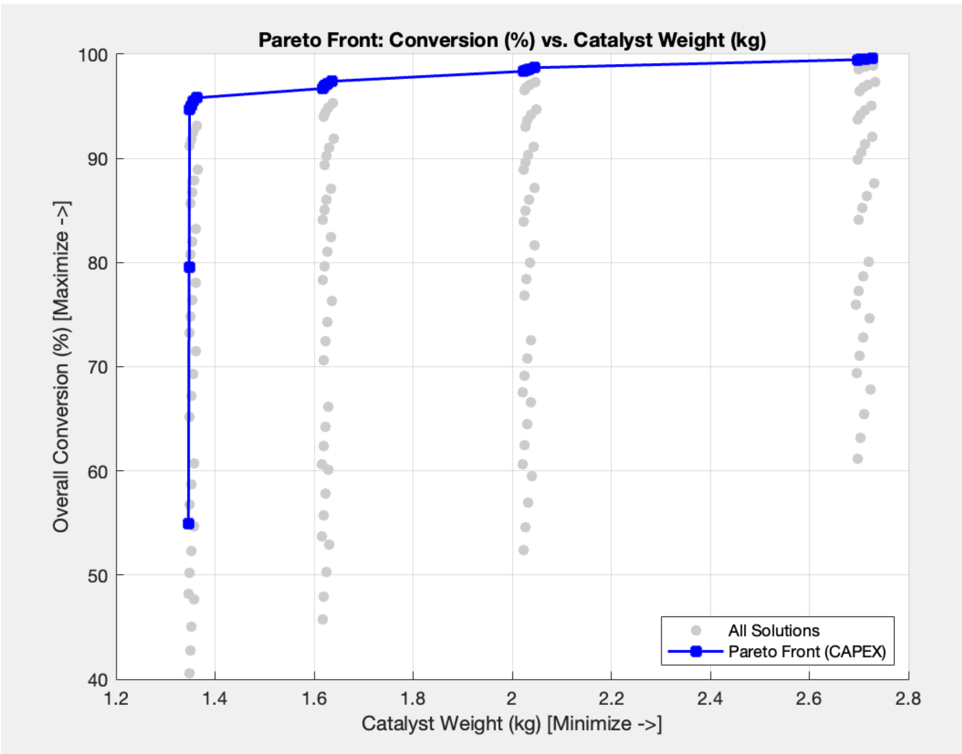


Figure 4: Pareto Front: Conversion vs. Catalyst Weight (CAPEX Analysis).

### 7.2.2 Operating Cost (OPEX) vs. Conversion Trade-off

This plot (Figure 5) maps the primary operational trade-off, revealing a highly non-linear cost curve:

- **The "Knee" (Best Value)**: The Pareto front has a sharp "knee" in the bottom-left, where conversion jumps from $\sim 40\%$ to $\sim 80\%$ with minimal power increase ($\sim 400$ kW to 402 kW). This region represents the most energy-efficient use of power.
- **Diminishing Returns**: Designs with conversion $>\sim 80\%$ exhibit clear diminishing returns. For example, moving from 80% to 84% conversion requires a substantial additional power penalty ($\sim 28$ kW), confirming the higher costs associated with aggressive conversion targets.

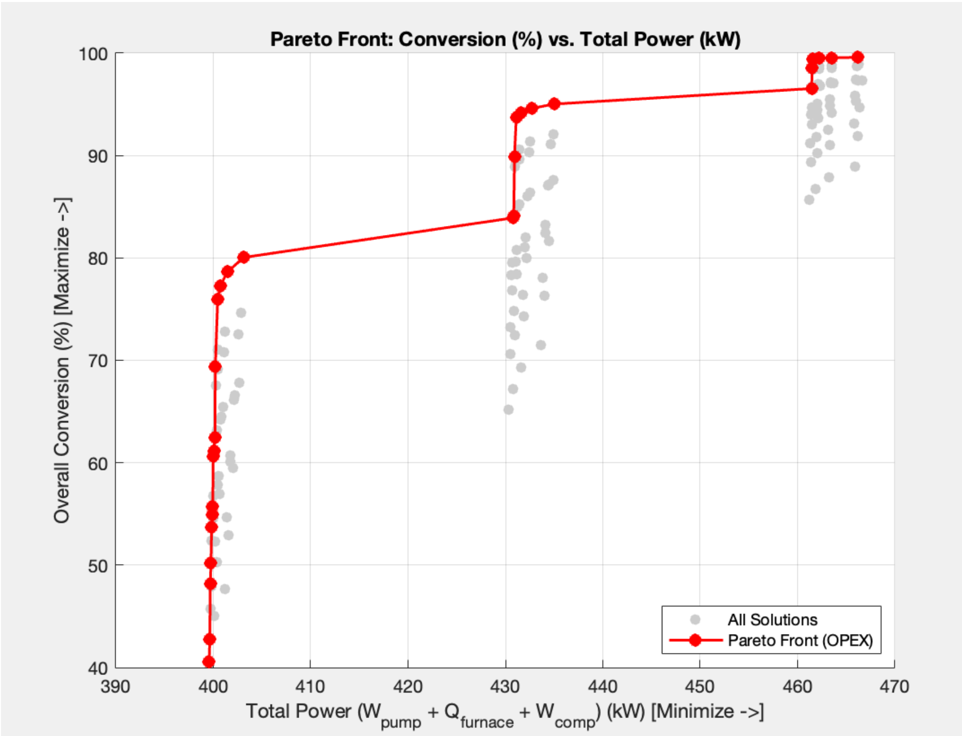


Figure 5: Pareto Front: Conversion vs. Total Power (OPEX Analysis).

## 7.3 Consolidated Process Efficiency Analysis

The custom Process Efficiency metric ($\eta_{process}$) was used as the single consolidated metric, combining OPEX and $H_2$ costs against sulfur removal, to find the true overall optimum.

1. **Efficiency vs. Operating Cost (OPEX)**: This plot (Figure 6) confirms the OPEX analysis. The "knee" is visible at $\sim 402$ kW, representing the "best value" design with the highest possible efficiency ($\sim 2.5 \times 10^{-4}$) at the lowest power cost. The front shows a poor operational range between 402 kW and 430 kW where efficiency is flat despite increased power.
2. **Efficiency vs. Conversion (The Final Result)**: This plot (Figure 7) is the most critical, answering the question of the most efficient level of conversion.

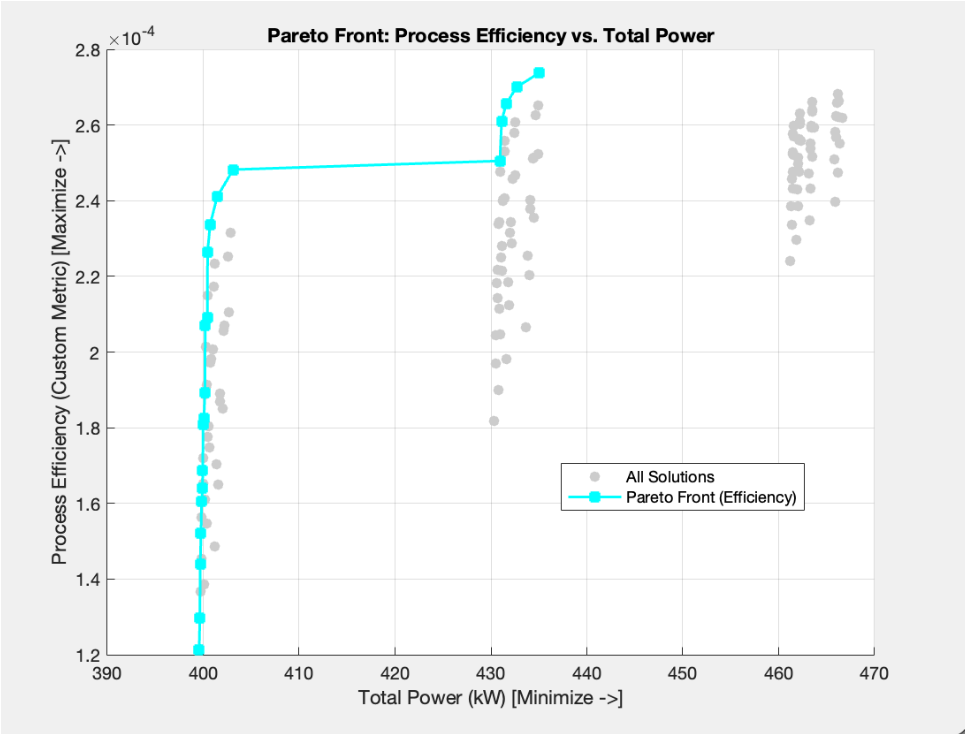


Figure 6: Pareto Front: Process Efficiency vs. Total Power.

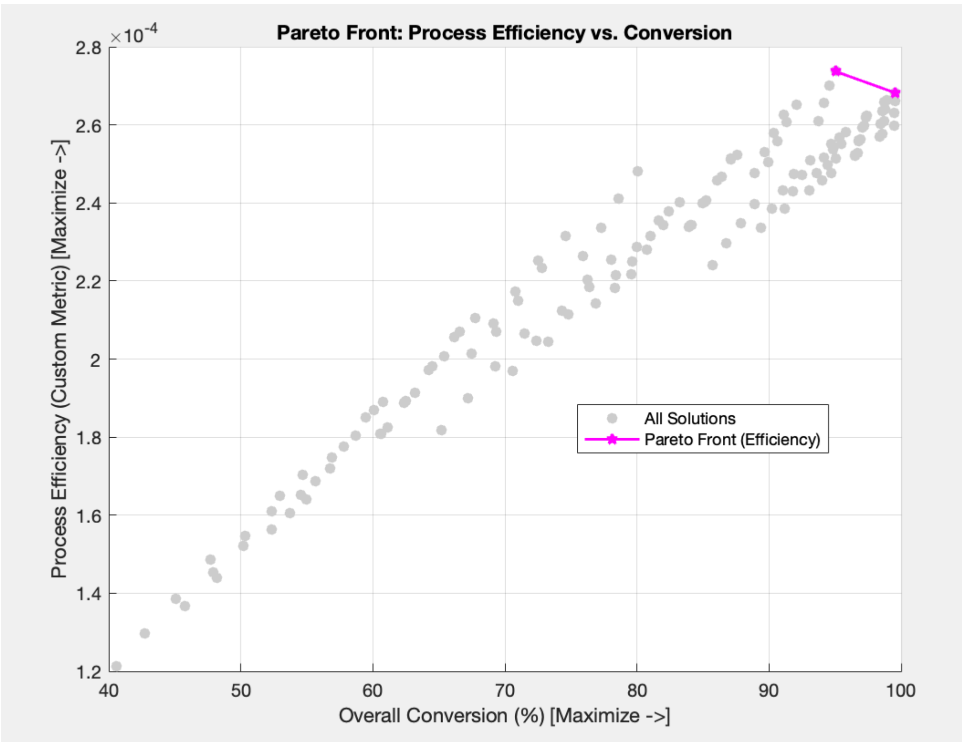


Figure 7: Pareto Front: Process Efficiency vs. Overall Conversion (The Final Design Criterion).

## 7.4 Summary of Optimal Strategy

The Pareto analysis provides definitive strategies for selecting the final design:

- **Peak Efficiency Strategy**: The overall most efficient design, per Equation 23, occurs at the peak of the efficiency-vs-conversion curve (Figure 7), at approximately 94% overall conversion ($\eta_{process} \approx 2.75 \times 10^{-4}$). The curve's subsequent backward "Hook" proves that exceeding 94% is sub-optimal due to excessive energy cost.

- **Best Value (Lowest OPEX) Strategy**: A design in the range of $\sim 80\%$ conversion and $\sim 402$ kW power input offers the best balance of performance for the lowest operating cost.
- **Capital-Efficient Strategy**: Designs operating at LHSV = 3.0 are the most capital-efficient.

# 8 Appendix: Key Methods Explained

## 8.1 Pareto Analysis

Pareto analysis is a multi-objective optimization technique used to find optimal solutions in the presence of trade-offs between conflicting objectives. The "Pareto front" is the set of all non-dominated solutions, where it is impossible to improve one objective without sacrificing performance in another.

## 8.2 Chao-Seader Thermodynamic Model

The Chao-Seader model is a semi-empirical method for predicting VLE K-values. The K-value ($K_i$) is defined as:

$$K_i = \frac{y_i}{x_i} = \frac{\nu_i \phi_i^{sat}}{\hat{\phi}_i} \tag{23}$$

Where $\nu_i$ is the liquid activity coefficient (from regular solution theory) and $\hat{\phi}_i$ is the vapor-phase fugacity coefficient (from the Redlich-Kwong equation of state).

## 8.3 Rachford-Rice Equation

The Rachford-Rice method is an iterative algorithm for solving flash calculations. It solves for the vapor fraction ($\psi = V/F$) by finding the root of the objective function $g(\psi)$:

$$g(\psi) = \sum_{i=1}^{N_c} \frac{z_i(K_i - 1)}{1 + \psi(K_i - 1)} = 0 \tag{24}$$

Once $\psi$ is found, the liquid ($x_i$) and vapor ($y_i$) compositions are calculated as:

$$x_i = \frac{z_i}{1 + \psi(K_i - 1)}, \quad y_i = K_i x_i \tag{25}$$

# References

[1] Buitrago, J., Amaya, D., & Ramos, O. (2017). Model and Simulation of a Hydrotreatment Reactor for Diesel Hydrodesulfurization in Oil Refining. *Contemporary Engineering Sciences, Vol. 10, no. 25*, 1245-1254. (online)